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(57) Abstract :

Disclosed is a green and cost-effective hydrothermal method for producing graphene quantum dots (GQDs). The invention utilizes benzoic resin as a natural carbon precursor, with citric acid as a stabilizing agent and cobalt acetate as a dopant and auxiliary carbon source, all combined in a one-pot reaction. The precursor mixture is heated at about 180 °C for 12 hours in a sealed autoclave, yielding a stable brownish colloidal solution of water-dispersible GQDs. The resulting graphene quantum dots are characterized by a conjugated sp² carbon core (evidenced by a prominent ~270 nm UV absorption and Raman D/G bands) and abundant surface functional groups from citric acid (confirmed by FTIR), as well as slight cobalt doping. XRD analysis shows a broad graphitic (002) peak ~24°, indicating the nanocrystalline nature of the GQDs. The process avoids toxic reagents and harsh conditions, making it environmentally friendly and scalable. The GQDs produced have excellent photoluminescence, stability, and biocompatibility, making them well-suited for applications in optoelectronics, bioimaging, sensing, and energy devices.

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